

Spatial Capacity of Random-Access Poisson Networks

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Abstract—In Internet-of-Things (IoT) networks with a massive number of nodes, severe interference can cause significant performance degradation if the access parameters are not properly configured. This paper addresses this issue by characterizing the spatial capacity of a slotted Aloha-based Poisson bipolar network under a general state-dependent backoff protocol. Adopting the saturated-traffic assumption, we develop a spatiotemporal framework that jointly captures the spatial interference among transmitters and their state-dependent channel-access behavior. By modeling the head-of-line (HOL) packet state transitions as a discrete-time Markov chain and invoking a decoupling approximation for the per-state transmitter processes, we derive a fixed-point expression for the steady-state probability of successful transmission and a tractable expression for the spatial capacity. On this basis, the maximum spatial capacity and the corresponding optimal initial transmission probability and receiver threshold are characterized in tractable semi-closed form: the optimal thresholds are determined by scalar fixed-point equations and admit closed-form expressions in the low- and high-SNR regimes. The accuracy of the analysis is verified through Monte-Carlo simulations. Our results reveal that, in the low-SNR regime, the maximum spatial capacity increases linearly with the received SNR while the optimal receiver threshold decreases as the node density grows; in the high-SNR regime, the network becomes interference-limited, and the optimal receiver threshold is determined solely by the path-loss exponent, independently of the node density. Furthermore, the maximum spatial capacity decreases monotonically as the transmitter-receiver distance increases.

Index Terms—Stochastic geometry, transmission protocol, slotted Aloha, capture model.

I. INTRODUCTION

The Internet of Things (IoT) is a network of nodes that gather information from the environment, spanning diverse application domains such as industrial automation, smart city infrastructure, environmental monitoring, and vehicular communications [1], [2]. In such networks, a massive number of low-complexity devices communicate with each other over shared wireless channels. Due to the heavy signaling overhead of centralized scheduling, distributed random-access schemes are usually adopted in large-scale IoT networks. With one of the most representative random-access schemes, slotted Aloha [3], [4], each node accesses the channel independently with a certain transmission probability. If the transmission probability is poorly configured, severe interference amongst nodes would be incurred due to the uncoordinated nature of transmitters, leading to significant performance degradation. Therefore, quantifying the fundamental performance limits of

such networks and deriving the optimal access parameters constitute problems of both theoretical and practical significance, and a satisfactory solution calls for rigorous performance analysis in the first place.

A. Spatiotemporal Modeling of Random-Access Networks

In IoT networks, each node is usually equipped with a buffer, and incoming packets form a queue in each node's buffer. The main difficulty associated with performance analysis comes from the interdependency among the evolution of different queues, which is usually referred to as interacting queues [5], [6]. In particular, because of the interference, the access behavior of one transmitter as well as its queue status can affect the transmission outcome of its neighbors, which in turn affects its own access behavior and queue status. Therefore, the service process of each transmitter's queue is coupled with each other, due to which exact analysis becomes prohibitively difficult. How to conduct tractable analysis has thus drawn significant attention. A consensus has gradually emerged that each node's queue can be approximately considered as an independent queueing system when the number of nodes is large, thereby decoupling the joint queue evolution into a set of tractable single-queue problems [7].

Stochastic geometry has emerged as the canonical mathematical framework for characterizing spatial randomness in wireless networks, enabling exact or tight approximate expressions for key performance metrics such as coverage probability and network throughput [8]–[10]. Under the standard assumption that transmitter locations follow a homogeneous Poisson point process (PPP), the aggregate interference can be expressed in closed form via Laplace-functional techniques, and the per-link signal-to-interference-plus-noise ratio (SINR) distribution becomes analytically tractable [3], [11].

To capture both the spatial and temporal dimensions of network behavior, a growing body of work has pursued spatiotemporal frameworks that jointly model queueing dynamics and spatial interference. Early work derived bounds on the SINR coverage probability by considering the two extremes of always-on transmitters and drop-upon-failure transmitters [12], [13]. Under light-traffic conditions, approximate expressions for coverage probability and stability regions were obtained in [14], [15]. The meta distribution was introduced in [16], [17] to provide a fine-grained characterization of per-link reliability beyond the spatially averaged success probability. A phase-type (PH) queue model was employed in [18] to encapsulate the queueing interactions via a set of fixed-point equations. More recent contributions have further refined the

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spatiotemporal analysis to accommodate heterogeneous traffic, non-Poisson topologies, and age-of-information metrics [19]–[21].

Despite these advances, a critical limitation persists in the majority of existing spatiotemporal models: the queueing interaction is captured solely through the *binary* queue state—i.e., whether a transmitter’s buffer is empty or not. In practice, however, the widely adopted *binary exponential backoff* (BEB) protocol [22] and its generalizations cause each transmitter to progressively reduce its channel access probability after successive failures, inducing a rich set of *state-dependent* access behaviors. This state-dependent interference structure is fundamentally different from the binary-state model and cannot be captured by existing frameworks without substantial modification. Accurately accounting for these state-dependent access probabilities is therefore essential for precise performance analysis and parameter optimization in practical random-access IoT networks.

This modeling approach has been applied to the analysis of Aloha [4], [7] and CSMA networks [23]. To isolate the impact of the state-dependent access behavior, the present work focuses on the *saturated-traffic spatial capacity*: we adopt the saturated-traffic assumption, i.e., each transmitter always has a packet in its buffer awaiting transmission. Under this assumption, the spatiotemporal framework characterizes the HOL packet state transitions induced by the backoff protocol, rather than the queue stability or the arrival-departure dynamics that arise under random packet arrivals. Consequently, the throughput metric studied in this paper is the saturated service rate of the typical link, and we do not claim any result on queueing stability; extending the framework to incorporate an explicit arrival process is left as future work.

B. Spatial Capacity: Characterization and Optimization

In the following, we briefly review past research on spatial capacity characterization and optimization in random-access networks. Gupta and Kumar [24] pioneered an important line of work on capacity in ad hoc networks, which measures the end-to-end throughput and terms it “transport capacity”, under the assumption that the network adopts the most efficient MAC and routing methods. In contrast, this paper emphasizes the network performance metric termed “spatial capacity”, which is defined as the average number of bits successfully delivered per unit time per unit area [25]. This metric provides a comprehensive perspective on the multiple access network by jointly considering the encoding rate of each packet and the overall network throughput, which encompasses aspects such as interference, signal strength, and the spatial distribution of nodes, and is therefore particularly suitable for network design and parameter optimization. The spatial capacity under path-loss-only and random fading channels has been studied in [25], the impact of packet retransmissions was analyzed in [26], and a spatial adaptation of Aloha within ad hoc networks was proposed in [27]. Significant research has also been dedicated to understanding the capacity limits of random-access systems incorporating the queueing state transition, yet without spatial modeling [4], [28]. However, for simplicity in

analysis, the spatial capacity in the aforementioned studies assumes that the nodes are perpetually active. Consequently, the sporadic nature of node access and the intricacies of the retransmission mechanism have been overlooked, thereby affecting the precise measurement of spatial capacity. The spatiotemporal correlation of interference in Aloha networks has been characterized in [29], which demonstrates the importance of such effects in performance analysis and retransmission mechanism design. However, under such static assumptions, deriving an explicit expression for the optimal spatial capacity under a dynamic retransmission protocol proves challenging. In this paper, we will fill this gap by combining the queueing state transition with spatial attributes, and investigate the limits of spatial capacity as well as the optimal decoding threshold and retransmission mechanism design.

C. Our Contributions

In this paper, we address the above gap by developing a unified spatiotemporal framework for the spatial capacity of a slotted Aloha-based Poisson bipolar network, under a general state-dependent backoff protocol. Each transmitter accesses the channel with probability $q_i = q_0 \cdot Q(i)$ after experiencing i consecutive transmission failures, where $Q(i)$ is an arbitrary monotone non-increasing function and q_0 is the initial transmission probability. By focusing on the typical transmitter-receiver pair and combining tools from stochastic geometry and Markov chain analysis, and invoking a decoupling (mean-field) approximation that treats the per-state transmitter processes as mutually independent Poisson point processes, we derive tractable analytical expressions for the steady-state probability of successful transmission, the spatial capacity, and the corresponding optimal parameters. This decoupling approximation is accurate in the regime of large, dense networks with high node mobility, and its accuracy is validated against Monte-Carlo simulations in Section V. Our main contributions are summarized as follows.

- Spatiotemporal analytical framework with state-dependent access. We establish a tractable spatiotemporal framework that explicitly models the state-dependent channel access behavior induced by a general backoff protocol. By characterizing the HOL packet state transition via a discrete Markov chain and mapping the steady-state distribution to the spatial density of transmitters in each backoff state, we derive a fixed-point equation for the steady-state probability of successful transmission p . The framework subsumes binary-state models as a special case ($K = 0$) and naturally incorporates channel interference, fading, additive noise, and the full backoff dynamics, enabling accurate performance assessment that is consistent with the result of existing work [3] in the limiting case.
- Spatial capacity and optimal transmission parameters. Building on the derived p , we obtain a tractable expression for the spatial capacity $C = \lambda \cdot \lambda_{\text{out}} \cdot \log_2(1 + \theta)$ and characterize the maximum throughput λ_{max} under two settings: an unconstrained per-state optimization over all access probabilities $\{q_i\}$, and the protocol-consistent

fixed backoff-shape optimization over the initial probability q_0 alone (with $q_i = q_0 Q(i)$). We show that the optimal initial probability is $q_0^* = \min\{q_m, 1\}$, and that $\lambda_{\max}$ is independent of the backoff function $\{Q(i)\}$ whenever $q_m \leq 1$ (high-density/high-SNR contention regime); when $q_m > 1$, the optimizer saturates at $q_0^* = 1$ and $\lambda_{\max}$ then depends on $\{Q(i)\}$. This provides directly actionable parameter settings for system design.

- Optimal receiver threshold and SNR-region analysis. We characterize the optimal receiver threshold θ_m that maximizes the spatial capacity $C_{\max}$. For a general SNR, θ_m is shown to be the root of a scalar fixed-point equation, while closed-form expressions are obtained in the low-SNR and high-SNR regimes. In the low-SNR regime, $C_{\max}$ increases *linearly* with ρ , the optimal θ_m decreases with node density λ , and $C_{\max}$ itself scales linearly with λ . In the high-SNR (interference-limited) regime, θ_m converges to a constant determined solely by the path-loss exponent α , independent of λ , and $C_{\max}$ saturates to a value determined by R and α alone. These SNR-region-specific insights provide clear engineering guidelines for threshold selection across a wide range of deployment scenarios.
- Impact of pair distance and backoff design. We prove that $C_{\max}$ is a monotone decreasing function of the pair distance R . We further clarify the role of the backoff design: when the initial transmission probability q_0 is optimized and the operating point falls in the contention-limited regime ($q_m \leq 1$), the maximum throughput $\lambda_{\max}$ is invariant to the specific backoff function $\{Q(i)\}$ and the cutoff phase K (Theorem 1); whereas under a *fixed, non-optimized* access probability q_0 —as is common in practical deployments—a larger cutoff phase K consistently enhances the spatial capacity, particularly in high-density regimes, by mitigating spatiotemporal interference through more conservative retransmissions. These two observations are complementary rather than contradictory, and together they provide concrete guidance for the joint design of the physical-layer decoding threshold, the MAC-layer backoff parameters, and the network deployment geometry.

The remainder of this paper is organized as follows. Section II presents the system model. Section III derives the steady-state probability of successful transmission of the typical link. Section IV characterizes and optimizes the maximum spatial capacity. Section V presents simulation results to validate the analysis. Section VI concludes the paper.

II. SYSTEM MODEL

Consider a discrete-time slotted Poisson bipolar network where the locations of the transmitters are distributed as a PPP $\Phi = \{x_i\}$ of intensity λ . Each transmitter x_i is paired with a receiver x_j at a fixed distance R and a random orientation. At the beginning of each time slot, the positions of the nodes are redrawn according to the high-mobility random-walk model [3], [10], under which the point process is re-realized as an independent PPP in every slot. This high-mobility assumption removes the temporal correlation of the

interference and, together with the mean-field decoupling approximation that treats the per-state transmitter processes as mutually independent PPPs, leads to a tractable PPP-based analysis. We emphasize that the spatial decoupling is an approximation rather than an exact consequence of the mobility model; its accuracy is assessed against Monte-Carlo simulations in Section V. Assume that each packet is sent with unit power, and occupies one time slot. For each pair (x_i, x_j) , the received power is then given by $h_{x_i, x_j} R^{-\alpha}$, where α is the path loss exponent and h_{x_i, x_j} is the fading coefficient. The fading coefficients of all the pairs remain static over one time slot, and are spatially and temporally independent. The mean received SNR ρ is then given by $\rho = \frac{R^{-\alpha}}{\sigma^2}$, where σ^2 is the additive noise power. Throughout this paper, we adopt the saturated traffic assumption: each transmitter always has at least one packet in its buffer awaiting transmission. This assumption is standard in the analysis of spatial capacity [25], [26] and allows us to focus on the steady-state behavior of the HOL packet state transition process induced by the backoff protocol, without modeling the packet arrival process.

In the analysis, we condition on the point x_0 at which the typical transmitter under consideration is located, where $R = |x_0|$ is the distance from this point to the origin, at which the corresponding typical receiver is located. According to Slivnyak's theorem, the law of $\Phi \setminus \{x_0\}$ conditioned on the fact that Φ has a point at x_0 is the same as the law of Φ . For the typical transmitter, its access behavior evolves over time. In particular, it sends a fresh Head-of-Line (HOL) packet with an initial transmission probability q_0 . We assume perfect and instantaneous feedback of the transmission outcome from the receiver. After the i th transmission failure, the typical transmitter transmits the HOL packet with probability q_i . To alleviate channel contention, $\{q_i\}_{i=0, \dots, K}$ is usually assumed to be a monotonically non-increasing sequence. Without loss of generality, we let $q_i = q_0 \cdot Q(i)$, where q_0 is the initial transmission probability and $Q(i)$ is an arbitrary monotonically non-increasing function of i with $Q(0) = 1$ and $Q(i) \leq Q(i-1)$ for $i = 1, \dots, K$.

The state transition process of each HOL packet of the typical transmitter can then be characterized by a discrete-time Markov process. As Fig. 1 shows, a fresh HOL packet is initially in State T, and moves to State 0 if it is not transmitted. A State- i HOL packet stays in State i if it is not transmitted. Here i denotes the number of collisions it experiences. Otherwise, it moves to State T if its transmission is successful, or State $\min(K, i+1)$ if the transmission fails, where K denotes the cutoff phase. Note that the cutoff phase K can be any non-negative integer. In Fig. 1, p_t represents the probability of successful transmission of HOL packets at time slot t . As the transmitted HOL packet in time slot t is successfully received if and only if the received SINR is larger than the receiver threshold θ , we have

$$p_t = \Pr\{\text{SINR}_t \geq \theta\}. \quad (1)$$

In this paper, we are interested in the spatial capacity, which is defined as the average received information rate per area,

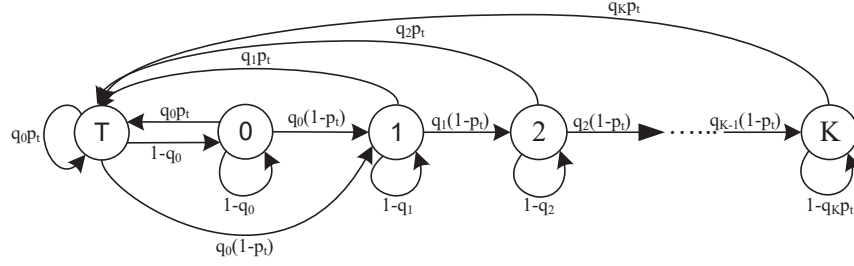


Fig. 1. Discrete Markov chain $\{X_j\}$ of the state transition process of the HOL packet of the typical transmitter.

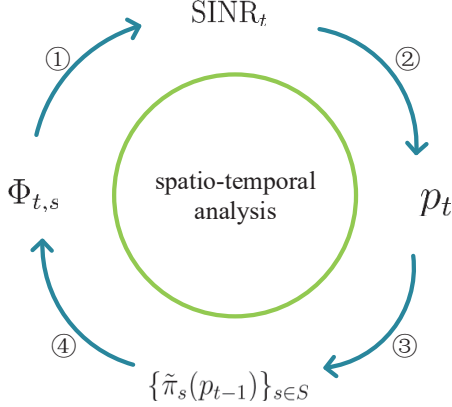


Fig. 2. The proposed spatiotemporal analysis framework.

is given by

$$C = \lambda \cdot \lambda_{\text{out}} \cdot \log_2(1 + \theta), \quad (2)$$

where λ_{out} denotes the throughput of the typical pair, i.e., the long-term average fraction of the channel time that is used for successful packet transmissions from the typical pair. Similar to [4], it is assumed that the transmitters are unaware of the instantaneous realizations of the small-scale fading coefficients. As a result, each transmitter independently encodes its information at a constant rate $\log_2(1 + \theta)$ bit/s/Hz.

III. PROBABILITY OF SUCCESSFUL TRANSMISSION OF HOL PACKETS

The spatial capacity performance of the Poisson bipolar network depends on the probability of successful transmission of each HOL packet, which will be first characterized in this section.

For the typical transmitter, whether its HOL packet is successfully transmitted or not depends on the interference. In the considered scenario, the interference at the typical receiver is determined not only by the spatial distribution of all the other transmitters, but also by the HOL packet states of all the other transmitters. Let $\Phi_{t,s}$ be the set of transmitters excluding x_0 that transmit a State- s HOL packet for time slot $t \in \mathbb{N}^+$, $s \in S = \{T, 0, 1, \dots, K\}$. The interference at the typical receiver located at the origin o at time slot t is then given

by

$$I_t = \sum_{s \in S} \sum_{x \in \Phi_{t,s}} h_x |x|^{-\alpha}. \quad (3)$$

Here h_x denotes $h_{x,o}$ for notational simplicity. The SINR of the typical receiver in time slot t is then given by

$$\text{SINR}_t = \frac{h_x R^{-\alpha}}{I_t + \sigma^2} = \frac{h_x}{\sum_{s \in S} \sum_{x \in \Phi_{t,s}} h_x |x|^{-\alpha} R^\alpha + 1/\rho}. \quad (4)$$

The probability of successful transmission of HOL packets at time slot t is given by

$$p_t = e^{-\theta/\rho} E[e^{-\theta R^\alpha I_t}], \quad (5)$$

by combining (1) and (4). To derive p_t , we resort to the Laplace transform of the interference at the typical receiver, which is given by

$$L_{I_t}(l) \triangleq \mathbb{E}[e^{-lI_t}] = \mathbb{E} \left[\prod_{s \in S} \prod_{x \in \Phi_{t,s}} \exp\{-lh_x |x|^{-\alpha}\} \right], \quad (6)$$

where the expectation is taken over both $\Phi_{t,s}$ and h_x .

Let $\pi_s(p_{t-1})$, $s \in S = \{T, 0, 1, \dots, K\}$, be the fraction of State- s transmitters at time slot $t-1$ over the whole Poisson bipolar network. $\Phi_{t,s}$ can then be viewed as a uniform PPP of intensity $\pi_s(p_{t-1})q_s\lambda$. Note that $q_T = q_0$. We then have

$$\begin{aligned} L_{I_t}(l) &\stackrel{(a)}{=} \prod_{s \in S} \mathbb{E} \left[\prod_{x \in \Phi_{t,s}} \exp\{-lh_x |x|^{-\alpha}\} \right] \\ &\stackrel{(b)}{=} \prod_{s \in S} \exp\{-2\pi\pi_s(p_{t-1})q_s\lambda \\ &\quad \times \int_0^\infty (1 - \mathbb{E}_{h_x}[\exp\{-lh_x |x|^{-\alpha}\}]) x dx\}, \end{aligned} \quad (7)$$

according to (6), where (a) follows from the decoupling (mean-field) approximation, under which the per-state transmitter processes $\{\Phi_{t,s}\}_{s \in S}$ are treated as mutually independent, and (b) follows from the fact that $\Phi_{t,s}$ can be viewed as a uniform PPP of intensity $\pi_s(p_{t-1})q_s\lambda$. This decoupling approximation, which is widely adopted in spatiotemporal analyses of random-access networks [18], [19], becomes asymptotically accurate as the network size and node mobility increase; its accuracy is corroborated by simulations in Section V. By combining (5-7), the probability of successful transmission of HOL packets at time slot t can be obtained as (8).

Under the high-mobility random-walk model, the SINR values of the typical receiver are independent and identically distributed (i.i.d.) across time slots t . Therefore, as $t \rightarrow \infty$, p_t converges to a steady-state value. Let $p = \lim_{t \rightarrow \infty} p_t$ denote the steady-state probability of successful transmission of HOL packets. Consequently, the Markov chain shown in Fig. 1 is uniformly strongly ergodic. Moreover, owing to the ergodicity of the PPP, the fraction of transmitters in State s at time slot t as $t \rightarrow \infty$ equals the probability that the typical transmitter is in State s . We then have

$$\lim_{t \rightarrow \infty} \pi_s(p_{t-1}) = \pi_s(\lim_{t \rightarrow \infty} p_{t-1}) = \pi_s(p), \quad (9)$$

where $\pi_s(p)$ is the steady-state distribution of the Markov chain, which is given by

$$\pi_T(p) = \frac{1}{\sum_{i=0}^{K-1} \frac{(1-p)^i}{q_i} + \frac{(1-p)^K}{pq_K}}, \quad (10)$$

and if $K = 0$

$$\pi_0(p) = \frac{1 - pq_0}{pq_0} \pi_T(p); \quad (11)$$

if $K \geq 1$

$$\begin{cases} \pi_0(p) = \frac{1-q_0}{q_0} \pi_T(p), \\ \pi_i(p) = \frac{(1-p)^i}{q_i} \pi_T(p), i = 1, \dots, K-1, \\ \pi_K(p) = \frac{(1-p)^K}{pq_K} \pi_T(p). \end{cases} \quad (12)$$

Finally, by combining (8-12), the steady-state probability of successful transmission of HOL packets is determined by the following fixed-point equation. It is clearly indicated in (13) that the steady-state probability of successful transmission of HOL packets of the typical pair is determined by parameters related to spatial distribution, such as the intensity λ and the transmitter-receiver distance R , and the transmission probabilities of the typical transmitter $\{q_i\}_{i=0,\dots,K}$ that evolve over time. Specifically, if h_x follows exponential distribution of unit mean, then p can be written as

$$p = e^{-\theta/\rho} \exp \left\{ -\frac{\lambda \pi \theta^{\frac{2}{\alpha}} R^2}{\text{sinc}(\frac{2\pi}{\alpha})} \cdot \frac{1}{\sum_{i=0}^{K-1} \frac{p(1-p)^i}{q_i} + \frac{(1-p)^K}{q_K}} \right\}, \quad (14)$$

where $\text{sinc}(\frac{2\pi}{\alpha}) = \sin(\frac{2\pi}{\alpha})/(\frac{2\pi}{\alpha})$. If the transmission probability of each transmitter remains constant over time, i.e., $q_s = q$, $\forall s \in \{T, 0, 1, \dots, K\}$, and there is no noise, i.e., $\rho \rightarrow \infty$, the probability of successful transmission can be expressed as

$$p = \exp \left\{ -\frac{\lambda \pi \theta^{\frac{2}{\alpha}} R^2}{\text{sinc}(\frac{2\pi}{\alpha})} \cdot q \right\}, \quad (15)$$

which is consistent with Eq. (3.4) in [3].

IV. MAXIMUM SPATIAL CAPACITY

In this section, we will derive the maximum spatial capacity. Specifically, it has been shown in Section II that the spatial capacity is determined by the information encoding rate $\log_2(1 + \theta)$ and the throughput of the typical pair λ_{out} . In the following, we first characterize the throughput of the typical pair λ_{out} .

In a saturated network, the throughput of the typical pair equals the service rate of the typical transmitter's queue, which is given by π_T since the queue of the typical transmitter has a successful output if and only if the HOL packet is in State T . We then have

$$\begin{aligned} \lambda_{\text{out}} &= \pi_T(p) \\ &= \frac{p(-\ln p - \theta/\rho)}{2\pi\lambda \int_0^\infty \left(1 - \mathbb{E}_{h_x} \left[\exp \left\{ -\theta R^\alpha h_x |x|^{-\alpha} \right\} \right] \right) x dx}, \end{aligned} \quad (16)$$

according to (13). It is clear from (16) that λ_{out} depends on p , which is in turn determined by the transmission probabilities of the typical transmitter $\{q_i\}_{i=0,\dots,K}$. Recall from the protocol definition $q_i = q_0 \mathcal{Q}(i)$ in Section II that the backoff shape $\mathcal{Q}(i)$ is fixed and only the initial transmission probability q_0 is tunable. We therefore distinguish two optimization settings:

- the *unconstrained per-state* setting, in which all access probabilities $\{q_i\}$ are treated as free design variables, yielding the benchmark $\lambda_{\text{max}}^{\text{uc}} = \max_{\{q_i\} \in [0,1]^{K+1}} \lambda_{\text{out}}$;
- the *fixed backoff-shape* setting, which respects the protocol constraint $q_i = q_0 \mathcal{Q}(i)$ and optimizes only the initial probability, i.e., $\lambda_{\text{max}} = \max_{q_0 \in [0,1]} \lambda_{\text{out}}$.

The following theorem characterizes both quantities and the corresponding optimal probabilities.

Theorem 1. *The maximum throughput of the typical pair under the unconstrained per-state setting, $\lambda_{\text{max}}^{\text{uc}} = \max_{\{q_i\}} \lambda_{\text{out}}$, is given by (17) which is achieved when the transmission probabilities of each node are set to be*

$$q_i = \begin{cases} q_m \mathcal{Q}(i) & \text{if } 2\pi\lambda \int_0^\infty \left(1 - \mathbb{E}_{h_x} \left[\exp \left\{ -\theta R^\alpha h_x |x|^{-\alpha} \right\} \right] \right) x dx \geq 1 \\ 1 & \text{otherwise,} \end{cases} \quad (18)$$

where q_m is given by

$$q_m = \frac{\sum_{i=0}^{K-1} \frac{\exp\{-1-\theta/\rho\}(1-\exp\{-1-\theta/\rho\})^i}{\mathcal{Q}(i)} + \frac{(1-\exp\{-1-\theta/\rho\})^K}{\mathcal{Q}(K)}}{2\pi\lambda \int_0^\infty \left(1 - \mathbb{E}_{h_x} \left[\exp \left\{ -\theta R^\alpha h_x |x|^{-\alpha} \right\} \right] \right) x dx}. \quad (19)$$

Under the fixed backoff-shape setting $q_i = q_0 \mathcal{Q}(i)$, the optimal initial transmission probability is

$$q_0^* = \min\{q_m, 1\}, \quad (20)$$

$$p_t = e^{-\theta/\rho} \prod_{s \in \mathcal{S}} \exp \left\{ -2\pi\pi_s(p_{t-1})q_s\lambda \int_0^\infty \left(1 - \mathbb{E}_{h_x} \left[\exp \left\{ -\theta R^\alpha h_x |x|^{-\alpha} \right\} \right] \right) x dx \right\}. \quad (8)$$

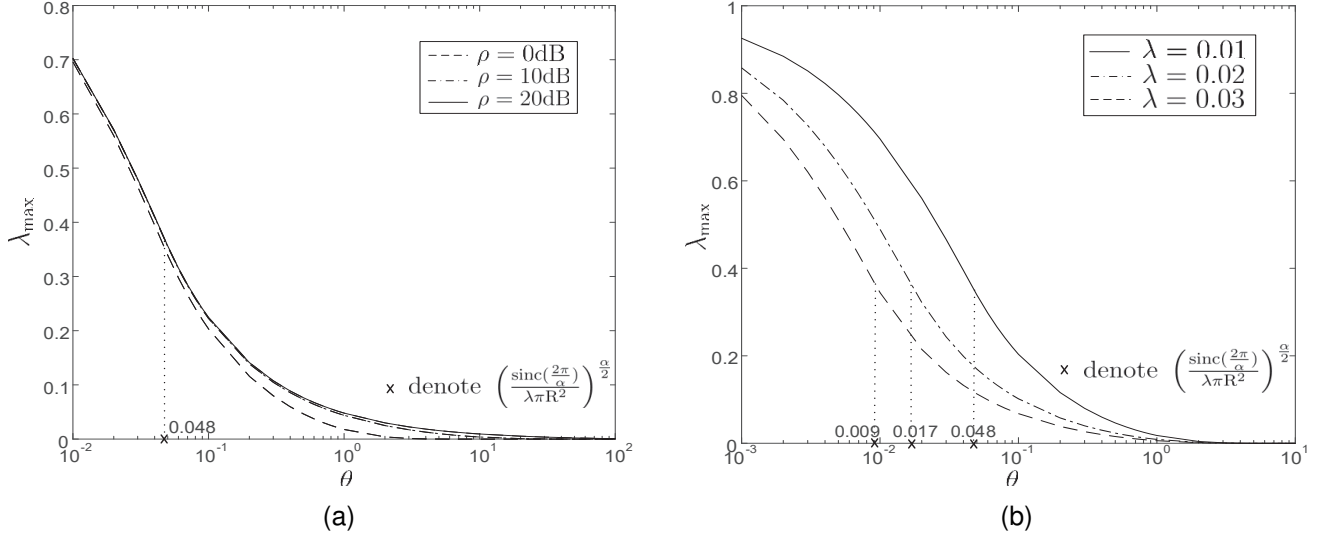


Fig. 3. The maximum throughput of the typical pair $\lambda_{\max}^{\text{uc}}$ (unconstrained per-state benchmark in Theorem 1, which is also attainable under the fixed backoff-shape protocol when $q_m \leq 1$): (a) $\lambda_{\max}^{\text{uc}}$ versus θ for various values of ρ , with $\lambda = 0.01$, $\alpha = 3$, $R = 10$; (b) $\lambda_{\max}^{\text{uc}}$ versus θ for various values of λ , with $\alpha = 3$, $R = 10$, $\rho = 0$ dB.

$$\begin{aligned}
 p &= e^{-\theta/\rho} \exp \left\{ -2\pi\lambda \int_0^\infty \left(1 - \mathbb{E}_{h_x} \left[\exp \left\{ -\theta R^\alpha h_x |x|^{-\alpha} \right\} \right] \right) x dx \cdot \frac{\pi_T(p)}{p} \right\} \\
 &= e^{-\theta/\rho} \exp \left\{ -2\pi\lambda \int_0^\infty \left(1 - \mathbb{E}_{h_x} \left[\exp \left\{ -\theta R^\alpha h_x |x|^{-\alpha} \right\} \right] \right) x dx \cdot \frac{1}{\sum_{i=0}^{K-1} \frac{p(1-p)^i}{q_i} + \frac{(1-p)^K}{q_K}} \right\}
 \end{aligned} \quad (13)$$

$$\lambda_{\max} = \begin{cases} \frac{\exp\{-1-\theta/\rho\}}{2\pi\lambda \int_0^\infty (1 - \mathbb{E}_{h_x} [\exp\{-\theta R^\alpha h_x |x|^{-\alpha}\}]) x dx} & \text{if } 2\pi\lambda \int_0^\infty (1 - \mathbb{E}_{h_x} [\exp\{-\theta R^\alpha h_x |x|^{-\alpha}\}]) x dx \geq 1 \\ \exp\{-\theta/\rho\} \cdot \exp \left\{ -2\pi\lambda \int_0^\infty (1 - \mathbb{E}_{h_x} [\exp\{-\theta R^\alpha h_x |x|^{-\alpha}\}]) x dx \right\} & \text{otherwise,} \end{cases} \quad (17)$$

and the resulting maximum throughput, denoted $\lambda_{\max}^{\mathcal{Q}}$, is

$$\lambda_{\max}^{\mathcal{Q}} = \begin{cases} \lambda_{\max}^{\text{uc}}, & q_m \leq 1, \\ \pi_T(p(q_0=1)), & q_m > 1, \end{cases} \quad (21)$$

where $p(q_0=1)$ denotes the success probability obtained from (13) with $q_0 = 1$. Throughout, $\lambda_{\max}^{\text{uc}}$ denotes the unconstrained per-state benchmark in (17) and $\lambda_{\max}^{\mathcal{Q}}$ denotes the throughput attainable under the protocol-consistent fixed backoff-shape setting; the two coincide whenever $q_m \leq 1$.

Proof: See Appendix A. ■

According to Theorem 1, if h_x follows exponential distribution of unit mean, then the maximum throughput of the typical pair $\lambda_{\max}$ can be written as

$$\lambda_{\max} = \begin{cases} \frac{\text{sinc}(\frac{2\pi}{\alpha})}{\lambda \pi \theta^{\frac{2}{\alpha}} R^2} \cdot \exp\{-1 - \frac{\theta}{\rho}\} & \text{if } \theta \geq \left(\frac{\text{sinc}(\frac{2\pi}{\alpha})}{\lambda \pi R^2} \right)^{\frac{\alpha}{2}} \\ \exp \left\{ -\frac{\lambda \pi \theta^{\frac{2}{\alpha}} R^2}{\text{sinc}(\frac{2\pi}{\alpha})} \right\} \cdot \exp\{-\frac{\theta}{\rho}\} & \text{otherwise,} \end{cases} \quad (22)$$

which is achieved when the transmission probabilities of each

node are set to be

$$q_i = \begin{cases} q_m \mathcal{Q}(i) & \text{if } \theta \geq \left(\frac{\text{sinc}(\frac{2\pi}{\alpha})}{\lambda \pi R^2} \right)^{\frac{\alpha}{2}} \\ 1 & \text{otherwise,} \end{cases} \quad (23)$$

where q_m is given by (24). It is clear from (17) that, under the unconstrained per-state setting, the maximum throughput of the typical pair $\lambda_{\max}^{\text{uc}}$ is independent of the backoff function $\{\mathcal{Q}(i)\}$, and is determined by the threshold θ , the received SNR ρ , the intensity λ and the transmitter-receiver distance R . Under the fixed backoff-shape setting, this independence carries over only when $q_m \leq 1$, i.e., when the interior optimizer $p^* = \exp\{-1 - \theta/\rho\}$ can be realized by a feasible initial access probability $q_0^* = q_m \leq 1$. Note that, since the numerator of q_m in (19) contains the factors $1/\mathcal{Q}(i)$, the condition $q_m \leq 1$ depends on the backoff shape $\{\mathcal{Q}(i)\}$ and is in general stronger than the unconstrained feasibility condition (the integral term in (17) being no smaller than one). When $q_m > 1$, the interior optimizer becomes infeasible under $q_i = q_0 \mathcal{Q}(i)$; the throughput is then maximized at the boundary $q_0^* = 1$, and the resulting $\lambda_{\max}^{\mathcal{Q}} = \pi_T(p(q_0=1))$ does depend on the backoff function $\{\mathcal{Q}(i)\}$ through (13). This regime typically arises at low node density or low SNR, where contention is mild and aggressive access is preferable. Fig. 3a

$$q_m = \frac{\text{sinc}(\frac{2\pi}{\alpha})}{\lambda\pi\theta^{\frac{2}{\alpha}}R^2} \cdot \left(\sum_{i=0}^{K-1} \frac{\exp\{-1 - \theta/\rho\}(1 - \exp\{-1 - \theta/\rho\})^i}{\mathcal{Q}(i)} + \frac{(1 - \exp\{-1 - \theta/\rho\})^K}{\mathcal{Q}(K)} \right). \quad (24)$$

demonstrates how the maximum throughput of the typical pair $\lambda_{\max}$ varies with the threshold θ with various values of the received SNR ρ . It can be observed that $\lambda_{\max}$ decreases as θ increases due to a lower probability of successful transmission of HOL packets, and increases as ρ increases. As $\rho \rightarrow \infty$, we have

$$\lim_{\rho \rightarrow \infty} \lambda_{\max} = \begin{cases} \frac{\text{sinc}(\frac{2\pi}{\alpha})}{\lambda\pi\theta^{\frac{2}{\alpha}}R^2} \cdot \exp\{-1\} & \text{if } \theta \geq \left(\frac{\text{sinc}(\frac{2\pi}{\alpha})}{\lambda\pi R^2} \right)^{\frac{\alpha}{2}} \\ \exp\left(-\frac{\lambda\pi\theta^{\frac{2}{\alpha}}R^2}{\text{sinc}(\frac{2\pi}{\alpha})} \right) & \text{otherwise.} \end{cases} \quad (25)$$

Moreover, it can be seen from Fig. 3b that $\lambda_{\max}$ decreases as λ increases since a higher spatial intensity of transmitters leads to more severe interference.

According to (2), the maximum spatial capacity under the unconstrained per-state benchmark is defined as

$$C_{\max}^{\text{uc}} = \max_{\{q_i\}, \theta} (\lambda \cdot \lambda_{\text{out}} \cdot \log_2(1 + \theta)). \quad (26)$$

By optimizing over $\{q_i\}$, λ_{out} is maximized at $\lambda_{\max}^{\text{uc}}$, which depends on the receiver threshold θ , according to (17). Therefore, θ can be further tuned to maximize the spatial capacity, i.e., we have

$$C_{\max}^{\text{uc}} = \max_{\theta} (\lambda \cdot \lambda_{\max}^{\text{uc}} \cdot \log_2(1 + \theta)). \quad (27)$$

The following theorem characterizes the maximum spatial capacity and the corresponding optimal receiver threshold θ_m under the unconstrained per-state benchmark when h_x follows an exponential distribution of unit mean. The same expression is attainable under the fixed backoff-shape protocol $q_i = q_0 \mathcal{Q}(i)$ whenever $q_m \leq 1$ (with $q_0^* = q_m$); when $q_m > 1$, the protocol-constrained optimum is attained at $q_0^* = 1$ and generally depends on the backoff shape $\{\mathcal{Q}(i)\}$. For brevity, we write $C_{\max}$ for $C_{\max}^{\text{uc}}$ in the remainder of this section.

Theorem 2. *The maximum spatial capacity $C_{\max}$ is given by*

$$C_{\max} = \begin{cases} \frac{\text{sinc}(\frac{2\pi}{\alpha})}{\pi R^2} e^{-1} \theta_h^{-\frac{2}{\alpha}} e^{-\frac{\theta_h}{\rho}} \log_2(1 + \theta_h) & \text{if } \rho \geq \rho_0 \\ \lambda e^{-\frac{\lambda\pi R^2}{\text{sinc}(\frac{2\pi}{\alpha})} \theta_l^{\frac{2}{\alpha}}} e^{-\frac{\theta_l}{\rho}} \log_2(1 + \theta_l) & \text{otherwise,} \end{cases} \quad (28)$$

where

$$\rho_0 = \frac{\left(\frac{\lambda\pi R^2}{\text{sinc}(\frac{2\pi}{\alpha})} \right)^{-\frac{\alpha}{2}}}{\frac{1}{\left(1 + \left(\frac{\lambda\pi R^2}{\text{sinc}(\frac{2\pi}{\alpha})} \right)^{\frac{\alpha}{2}} \right) \ln \left(1 + \left(\frac{\lambda\pi R^2}{\text{sinc}(\frac{2\pi}{\alpha})} \right)^{-\frac{\alpha}{2}} \right)} - \frac{2}{\alpha}}, \quad (29)$$

which is achieved when the receiver threshold θ is tuned to be

$$\theta = \theta_m = \begin{cases} \theta_h & \text{if } \rho \geq \rho_0 \\ \theta_l & \text{otherwise.} \end{cases} \quad (30)$$

θ_h is determined by

$$\frac{\theta}{1 + \theta} = \ln(1 + \theta) \cdot \left(\frac{2}{\alpha} + \frac{\theta}{\rho} \right), \quad (31)$$

when $\alpha \geq 2$. θ_l is determined by

$$\frac{\theta}{1 + \theta} = \ln(1 + \theta) \cdot \left(\frac{2}{\alpha} \cdot \frac{\lambda\pi\theta^{\frac{2}{\alpha}}R^2}{\text{sinc}(\frac{2\pi}{\alpha})} + \frac{\theta}{\rho} \right). \quad (32)$$

Under the fixed backoff-shape protocol $q_i = q_0 \mathcal{Q}(i)$, the corresponding initial transmission probability that achieves the maximum spatial capacity $C_{\max}$ is $q_0^* = \min\{q_m, 1\}$, with q_m given by (19).

Proof: See Appendix B. ■

A. Optimal receiver threshold θ_m

Theorem 2 indicates that to achieve the maximum spatial capacity, the receiver threshold θ should be carefully tuned. Fig. 4a illustrates how the optimal receiver threshold θ_m varies with mean received SNR ρ for a fixed R and α . As $\rho = \frac{R^{-\alpha}}{\sigma^2}$, ρ is solely determined by the noise power σ^2 and varies due to different values of σ^2 . It can be clearly observed from Fig. 4a that at the low SNR region, i.e., $\rho < \rho_0$, the optimal receiver threshold θ_m decreases as λ increases. As $\rho \rightarrow 0$, we have

$$\lim_{\rho \rightarrow 0} \theta_m = \lim_{\rho \rightarrow 0} \theta_l = \rho, \quad (33)$$

which linearly increases as ρ increases. At the high SNR region, i.e., $\rho \geq \rho_0$, in contrast, the optimal receiver threshold θ_m becomes independent of λ . As $\rho \rightarrow \infty$, we have

$$\lim_{\rho \rightarrow \infty} \theta_m = \lim_{\rho \rightarrow \infty} \theta_h = e^{\mathbb{W}_0\left(-\frac{\alpha}{2}e^{-\frac{\alpha}{2}}\right) + \frac{\alpha}{2}} - 1, \quad (34)$$

which is solely determined by the path loss exponent α . As Fig. 4a illustrated, (33) and (34) provide a good approximation for the optimal receiver threshold θ_m at the low SNR region and the high SNR region, respectively.

When the receiver threshold is optimally tuned, i.e., $\theta = \theta_m$, the maximum throughput of the typical pair can be further obtained as

$$\lambda_{\max}^{\theta=\theta_m} = \begin{cases} \frac{\text{sinc}(\frac{2\pi}{\alpha})}{\lambda\pi\theta_h^{\frac{2}{\alpha}}R^2} \cdot \exp\{-1 - \frac{\theta_h}{\rho}\} & \text{if } \rho \geq \rho_0 \\ \exp\left\{ -\frac{\lambda\pi\theta_l^{\frac{2}{\alpha}}R^2}{\text{sinc}(\frac{2\pi}{\alpha})} \right\} \cdot \exp\{-\frac{\theta_l}{\rho}\} & \text{otherwise,} \end{cases} \quad (35)$$

by combining (22) and Theorem 2. Fig. 4b illustrates how the maximum throughput $\lambda_{\max}^{\theta=\theta_m}$ varies with the mean received SNR ρ . As we can see from Fig. 4b, $\lambda_{\max}^{\theta=\theta_m}$ monotonically decreases with ρ due to a larger receiver threshold θ_m . Specifically, $\lambda_{\max}^{\theta=\theta_m}$ is upper-bounded by

$$\lim_{\rho \rightarrow 0} \lambda_{\max}^{\theta=\theta_m} \approx e^{-1} \quad (36)$$

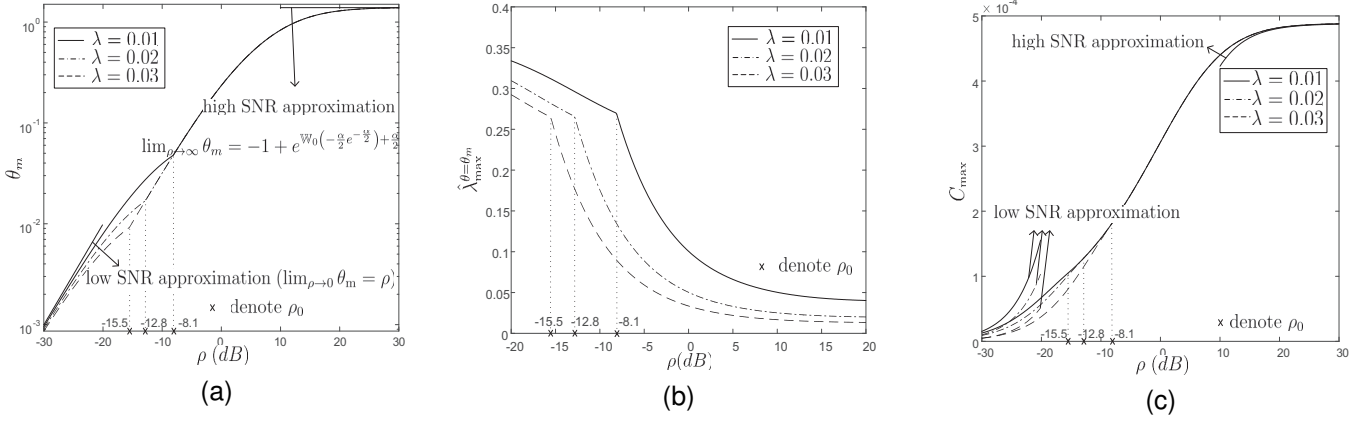


Fig. 4. Illustration of Theorem 2 and the SNR-region analysis: (a) the optimal receiver threshold θ_m versus ρ for various values of λ ; (b) the maximum throughput $\lambda_{\max}^{\theta=\theta_m}$ versus ρ for various values of λ ; and (c) the maximum spatial capacity $C_{\max}$ versus ρ for various values of λ . $\alpha = 3$, $R = 10$.

and lower-bounded by

$$\lim_{\rho \rightarrow \infty} \lambda_{\max}^{\theta=\theta_m} \approx \frac{\text{sinc}\left(\frac{2\pi}{\alpha}\right)}{\lambda \pi R^2} e^{-1} \cdot (e^{\mathbb{W}_0\left(-\frac{\alpha}{2}e^{-\frac{\alpha}{2}}\right) + \frac{\alpha}{2}} - 1)^{\frac{2}{\alpha}}, \quad (37)$$

which is solely determined by the intensity λ , the transmitter-receiver distance R and the path loss exponent α .

B. Maximum spatial capacity at different SNR regions

Let us take a closer look at the maximum spatial capacity $C_{\max}$ at different SNR regions. At the low SNR region, i.e., $\rho < \rho_0$, it has been shown in Section IV-A that the optimal receiver threshold θ_m approaches to ρ as $\rho \rightarrow 0$. The slope of the maximum spatial capacity in this case can then be written as

$$\lim_{\rho \rightarrow 0} \frac{C_{\max}}{\rho} = \lambda e^{-1} \log_2 e, \quad (38)$$

which indicates a linear increase with respect to ρ at the low SNR region, as illustrated in Fig. 4c. In this case, as the interference power is much smaller than that of the noise, the maximum spatial capacity $C_{\max}$ also linearly increases as the intensity λ increases.

On the other hand, at the high SNR region, i.e., $\rho \geq \rho_0$, it has been shown in Section IV-A that as $\rho \rightarrow \infty$, the optimal receiver threshold

$$\lim_{\rho \rightarrow \infty} \theta = \lim_{\rho \rightarrow \infty} \theta_h = e^{\mathbb{W}_0\left(-\frac{\alpha}{2}e^{-\frac{\alpha}{2}}\right) + \frac{\alpha}{2}} - 1. \quad (39)$$

The corresponding maximum spatial capacity can be written as

$$\begin{aligned} \lim_{\rho \rightarrow \infty} C_{\max} &= \frac{\text{sinc}\left(\frac{2\pi}{\alpha}\right)}{\pi R^2} e^{-1} \times (e^{\mathbb{W}_0\left(-\frac{\alpha}{2}e^{-\frac{\alpha}{2}}\right) + \frac{\alpha}{2}} - 1)^{\frac{2}{\alpha}} \\ &\quad \times \frac{\mathbb{W}_0\left(-\frac{\alpha}{2}e^{-\frac{\alpha}{2}}\right) + \frac{\alpha}{2}}{\ln 2}. \end{aligned} \quad (40)$$

In contrast to the low-SNR case, we can see that as $\rho \rightarrow \infty$, the maximum spatial capacity $C_{\max}$ is solely determined by the transmitter-receiver distance R and the path loss exponent α according to (40). In the high SNR region, the network becomes interference-limited, and thus $C_{\max}$ becomes independent of the intensity λ . As we can see from Fig. 4c, (40)

can be regarded as a good approximation for the maximum spatial capacity at the high SNR region.

C. Effect of pair distance on the maximum spatial capacity

In this section, the explicit expressions of the maximum spatial capacity $C_{\max}$ and the corresponding optimal receiver threshold θ are derived and given in (28) and (30-32), respectively. It can be clearly seen from (28) that $C_{\max}$ is a monotonic decreasing function of the pair distance R , as illustrated in Fig. 5b. Intuitively, a longer pair distance would result in performance degradation in terms of the SINR of the transmitted packet at the receiver side due to path loss, which reduces the probability of successful transmission of HOL packets.

V. SIMULATION RESULTS

In this section, Monte-Carlo simulation results are presented to verify the preceding analysis and to assess the accuracy of the decoupling approximation. The simulation is configured as follows. The transmitters are realized as an independent homogeneous PPP of intensity λ over a square region of $100 \times 100 \text{ m}^2$, and a toroidal (wrap-around) boundary correction is applied so that the finite window approximates an infinite PPP and edge effects are eliminated. We focus on the typical link, whose receiver is placed at the center of the region; its associated transmitter is located at distance R with a random orientation. At the beginning of each time slot, the locations of all transmitter-receiver pairs are independently redrawn according to the high-mobility random-walk model, while the typical link is kept fixed. For each transmitter, the binary exponential backoff is adopted, i.e., the transmission probability after the j th transmission failure is $q_i = q_0 \cdot 2^{-i}$ with $i = \min\{j, K\}$. Each transmitter starts from a fresh HOL packet (state T). To remove the influence of the transient phase, the first 2×10^3 time slots of every run are discarded as a warm-up period, and statistics are then collected over the subsequent 10^4 time slots; the reported curves are averaged over 10^3 independent Monte-Carlo runs (i.e., independent PPP realizations). The simulated statistics are collected at the typical pair; for instance, the simulated probability of

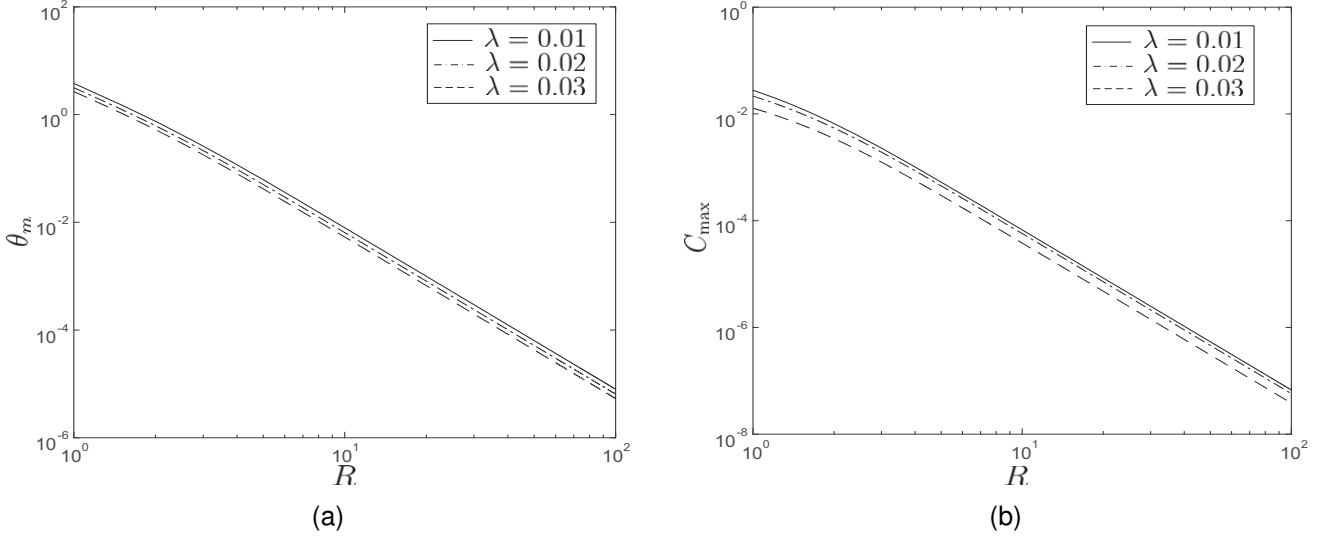


Fig. 5. Effect of the pair distance R , illustrating Theorem 2: (a) the optimal receiver threshold θ_m versus R for various values of λ ; and (b) the maximum spatial capacity $C_{\max}$ versus R for various values of λ , confirming that $C_{\max}$ is monotonically decreasing in R . $\sigma^2 = 0.1$, $\alpha = 3$.

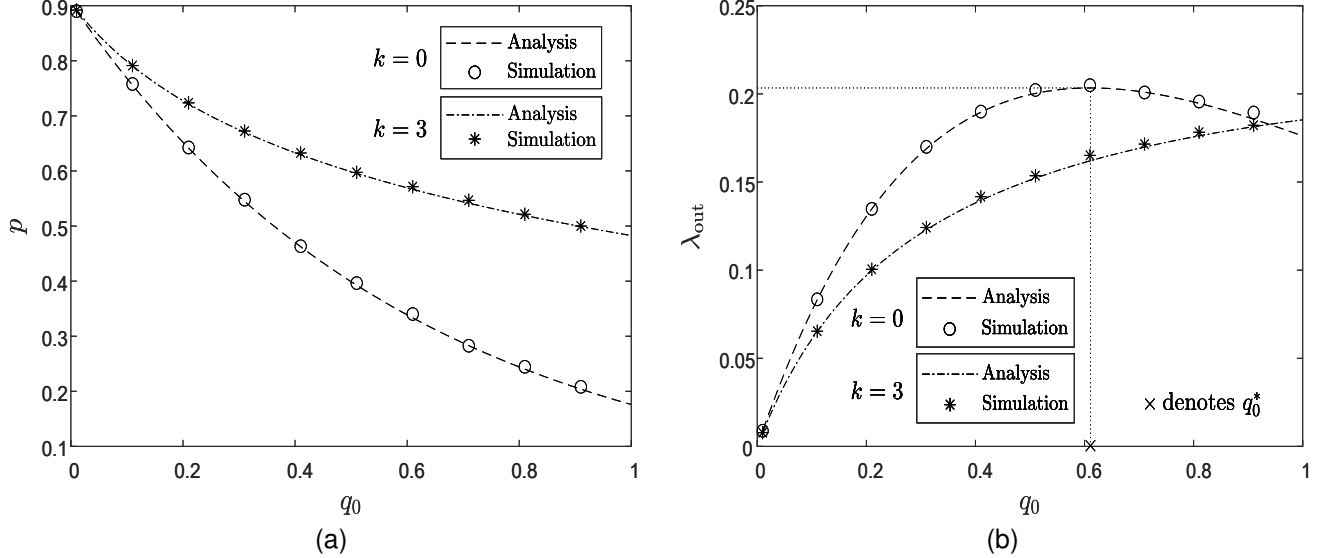


Fig. 6. (a) The probability of successful transmission of HOL packets p versus the initial transmission probability q_0 under various values of the cutoff phase K , validating the fixed-point characterization in (13); and (b) the throughput of the typical pair λ_{out} versus q_0 under various values of K , validating (16). $R = 10$, $\sigma^2 = 0.001$, $\alpha = 3$, $\lambda = 0.01$, $\theta = 0.1$, $\mathcal{Q}(i) = 2^{-i}$.

successful transmission of HOL packets is computed as the ratio of the number of successfully transmitted packets to the number of transmission attempts of the typical pair. Across all figures, the gap between the analytical and simulated curves is within a few percent and the corresponding 95% confidence intervals are smaller than the marker size; they are therefore omitted from the plots for clarity. The remaining parameters follow the system model in Section II.

The probability of successful transmission of HOL packets p has been characterized in Section III. Fig. 6a illustrates both the analytical and simulation results of p versus the initial transmission probability q_0 under various values of the cutoff phase K . It is observed that p decreases monotonically with q_0 , since a higher transmission probability intensifies the ag-

gregate interference from concurrent transmitters. In contrast, p increases with K , as a larger cutoff phase alleviates channel contention through a more conservative retransmission policy. The throughput of the typical pair λ_{out} has been derived in (16), and the corresponding performance is illustrated in Fig. 6b. It is shown that with $K = 0$, λ_{out} is maximized at an optimal $q_0 = 0.61$, reflecting a fundamental tradeoff between channel access opportunity and interference level. With $K = 3$, in contrast, λ_{out} increases monotonically with q_0 and attains its maximum at $q_0 = 1$, since the backoff mechanism effectively suppresses collisions at high transmission probabilities. The close agreement between analytical and simulation results validates the accuracy of the proposed spatiotemporal framework.

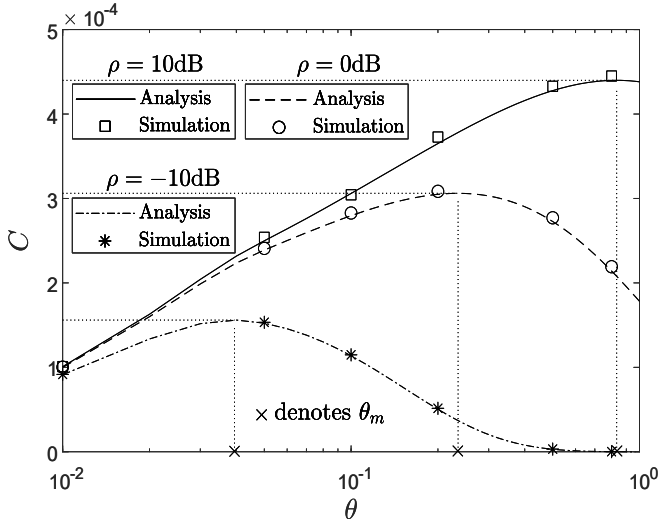


Fig. 7. The spatial capacity C versus the receiver threshold θ under various values of the mean received SNR ρ , validating the optimal receiver threshold θ_m characterized in Theorem 2. The markers denote the optimal threshold θ_m predicted by (30). $\lambda = 0.01$, $q_0 = q_0^*$, $K = 0$, $\sigma^2 = 0.001$, $\alpha = 3$, $R = 10$.

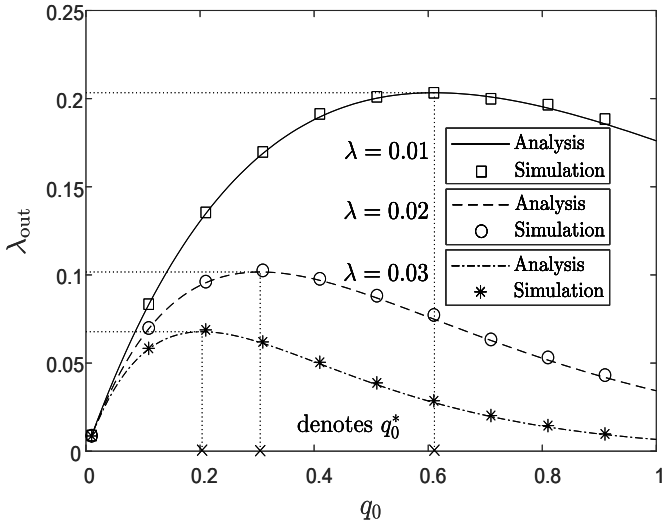


Fig. 8. The throughput of the typical pair λ_{out} versus the initial transmission probability q_0 under various values of the spatial intensity λ , illustrating the existence of an optimal q_0 consistent with Theorem 1. $\theta = 0.1$, $R = 10$, $\sigma^2 = 0.001$, $\alpha = 3$, $\mathcal{Q}(i) = 2^{-i}$.

Section IV further demonstrates that the receiver threshold θ must be carefully tuned to maximize the spatial capacity C . Fig. 7 illustrates the variation of C with respect to θ when the throughput of the typical pair λ_{out} is maximized, i.e., q_0 is configured according to (19). It is verified that there exists an optimal receiver threshold θ_m that maximizes C , and this optimal value is governed by the mean received SNR ρ , which is consistent with the analytical results in Theorem 2.

Fig. 8 investigates the effect of the spatial intensity λ on the throughput of the typical pair λ_{out} . It is observed that for each value of λ , λ_{out} first increases with q_0 and then decreases after attaining a maximum, indicating the existence of an optimal transmission probability q_0^* . This unimodal behavior stems from the inherent tradeoff between channel utilization

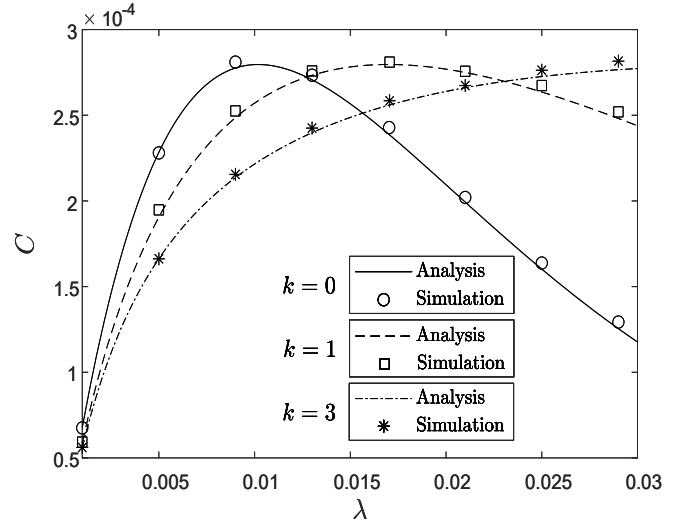


Fig. 9. The spatial capacity C versus the spatial intensity λ under various values of the cutoff phase K , for a fixed, non-optimized initial transmission probability $q_0 = 0.6$. Under this practical setting, a larger K improves C , especially at high density. $q_0 = 0.6$, $\theta = 0.1$, $R = 10$, $\sigma^2 = 0.001$, $\alpha = 3$, $\mathcal{Q}(i) = 2^{-i}$.

and interference: a small q_0 leads to underutilization of the channel, whereas a large q_0 results in excessive interference that degrades the probability of successful transmission. Furthermore, as λ increases, the maximum achievable throughput decreases and q_0^* shifts toward smaller values, since a higher node density intensifies the aggregate interference and necessitates a more conservative channel access strategy.

The spatial capacity C as a function of the spatial intensity λ is further examined in Fig. 9 under different cutoff phases K . It is observed that C first increases with λ and then decreases after reaching a peak, revealing the existence of an optimal network density. At low λ , the spatial reuse gain dominates, so deploying more transmitters improves C . As λ continues to grow, however, the resulting interference severely degrades the probability of successful transmission, leading to a reduction in spatial capacity. Moreover, for the fixed, non-optimized access probability $q_0 = 0.6$ adopted here, a larger cutoff phase K consistently yields a higher spatial capacity across all values of λ , confirming that under such practical settings a more flexible backoff scheme effectively manages spatiotemporal interference and enhances overall network efficiency. We emphasize that this gain pertains to the non-optimized regime; once q_0 is optimally tuned and the operating point lies in the contention-limited regime ($q_m \leq 1$), the maximum throughput becomes invariant to K , as established in Theorem 1. The simulation results are in close agreement with the analytical curves, further validating the theoretical analysis in Section IV.

VI. CONCLUSION

In this paper, a spatiotemporal framework is proposed to optimize the spatial capacity in a slotted Aloha-based Poisson bipolar network. By considering the spatiotemporal interactions and state-dependent access behavior amongst the

transmitters, a fixed-point equation for the steady-state probability of successful transmission of HOL packets is derived, based on which the optimal spatial capacity, the optimal initial transmission probability $q_0^* = \min\{q_m, 1\}$, and the optimal receiver threshold θ_m are further characterized under a tractable decoupling approximation. We find that, once the initial transmission probability q_0 is optimized, the maximum throughput $\lambda_{\max}$ is independent of the specific backoff function $\{Q(i)\}$ and the cutoff phase K in the contention-limited regime ($q_m \leq 1$), where it is solely determined by θ , ρ , λ , and R ; when $q_m > 1$, the optimal initial probability saturates at $q_0^* = 1$ and $\lambda_{\max}$ does depend on $\{Q(i)\}$. Besides, we find that: 1) At the low SNR region, $C_{\max}$ and θ_m both scale linearly with ρ , and $C_{\max}$ increases linearly with λ . 2) At the high SNR region, the network becomes interference-limited, θ_m converges to a constant determined solely by the path loss exponent α , and $C_{\max}$ saturates to a value independent of λ . 3) $C_{\max}$ is a monotone decreasing function of the pair distance R . 4) Under a fixed, non-optimized access probability q_0 , a larger cutoff phase K consistently enhances the spatial capacity, particularly in high-density regimes; this is complementary to the K -invariance of $\lambda_{\max}$ under optimized q_0 stated above. The accuracy of the analytical framework is verified via simulations, and the analytical results are shown to be in close agreement with the simulation results throughout.

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APPENDIX A PROOF OF THEOREM 1

According to (13) and (16), we have $\lambda_{\max} = \max_{\{p\}} \lambda_{\text{out}}$ with the constraint (41).

By letting $\frac{d\lambda_{\text{out}}}{dp} = 0$, we have

$$p = \exp\{-1 - \theta/\rho\}. \quad (42)$$

Therefore, if

$$2\pi\lambda \int_0^\infty \left(1 - \mathbb{E}_{h_x} \left[\exp \left\{ -\theta R^\alpha h_x |x|^{-\alpha} \right\} \right] \right) x dx \geq 1, \quad (43)$$

then λ_{out} is maximized at

$$p = \exp\{-1 - \theta/\rho\}; \quad (44)$$

$$p \in \left[\exp\{-\theta/\rho\} \cdot \exp\left\{-2\pi\lambda \int_0^\infty \left(1 - \mathbb{E}_{h_x} \left[\exp\left\{-\theta R^\alpha h_x |x|^{-\alpha}\right\}\right] \right) x dx \right\}, \exp\{-\theta/\rho\} \right]. \quad (41)$$

otherwise, λ_{out} , is maximized at

$$p = \exp\{-\theta/\rho\} \cdot \exp\left\{-2\pi\lambda \int_0^\infty \left(1 - \mathbb{E}_{h_x} \left[\exp\left\{-\theta R^\alpha h_x |x|^{-\alpha}\right\}\right] \right) x dx \right\}. \quad (45)$$

The corresponding optimal transmission probabilities can be obtained according to (13). Under the unconstrained per-state setting, any feasible p in (41) can be realized by a suitable choice of $\{q_i\} \in [0, 1]^{K+1}$; in particular, the lower boundary of (41) (corresponding to maximum interference) is attained at $q_i = 1$ for all i , which gives the second branch of (17), whereas the upper boundary $p = \exp\{-\theta/\rho\}$ is approached as the access probabilities tend to zero. Under the fixed backoff-shape constraint $q_i = q_0 \mathcal{Q}(i)$, however, the access probabilities are coupled through q_0 . Substituting $q_i = q_0 \mathcal{Q}(i)$ into (13) shows that the interior optimizer $p^* = \exp\{-1-\theta/\rho\}$ is attained at $q_0 = q_m$ given by (19); this is feasible if and only if $q_m \leq 1$, in which case $\lambda_{\text{max}} = \lambda_{\text{max}}^{\text{uc}}$ and is independent of $\{\mathcal{Q}(i)\}$. When $q_m > 1$, the optimizer is infeasible and, since $\lambda_{\text{out}} = \pi_T(p)$ is monotonically increasing in p over the feasible range below p^* , the maximum is attained at the boundary $q_0^* = 1$, yielding $\lambda_{\text{max}} = \pi_T(p(q_0=1))$, which depends on $\{\mathcal{Q}(i)\}$ through (13).

APPENDIX B PROOF OF THEOREM 2

Throughout this proof, we retain the noise power σ^2 for algebraic convenience; it is related to the mean received SNR ρ used in the main text through $\rho = R^{-\alpha}/\sigma^2$, so that $R^\alpha \sigma^2 = 1/\rho$. The two notations are therefore interchangeable, and all results below can be equivalently expressed in terms of ρ . Let $x = \frac{\lambda \pi R^{\frac{2}{\alpha}}}{\text{sinc}(\frac{2\pi}{\alpha})}$. According to (22) and (27), the maximum spatial capacity C_{max} can be written as $C_{\text{max}} = \max_{\theta > 0} f(\theta)$, where

$$f(\theta) = \begin{cases} \frac{\lambda}{x} e^{-1} \cdot \theta^{-\frac{2}{\alpha}} e^{-R^\alpha \sigma^2 \theta} \log_2(1 + \theta) & \text{if } \theta \geq x^{-\frac{\alpha}{2}}, \\ \lambda e^{-x\theta^{\frac{2}{\alpha}}} e^{-R^\alpha \sigma^2 \theta} \log_2(1 + \theta) & \text{otherwise.} \end{cases} \quad (46)$$

Let

$$f_1(\theta) = \frac{\lambda}{x} e^{-1} \cdot \theta^{-\frac{2}{\alpha}} e^{-R^\alpha \sigma^2 \theta} \log_2(1 + \theta) \quad (47)$$

and

$$f_2(\theta) = \lambda e^{-x\theta^{\frac{2}{\alpha}}} e^{-R^\alpha \sigma^2 \theta} \log_2(1 + \theta). \quad (48)$$

Note that $f_1(x^{-\frac{\alpha}{2}}) = f_2(x^{-\frac{\alpha}{2}})$. Accordingly, we have $C_{\text{max}} = \max(C_1, C_2)$, where $C_1 = \max_{\theta \geq x^{-\frac{\alpha}{2}}} f_1(\theta)$ and $C_2 = \max_{\theta \leq x^{-\frac{\alpha}{2}}} f_2(\theta)$.

By solving $C_1 = \max_{\theta \geq x^{-\frac{\alpha}{2}}} f_1(\theta)$ and $C_2 = \max_{\theta \leq x^{-\frac{\alpha}{2}}} f_2(\theta)$, we

have

$$C_1 = \begin{cases} f_1(\theta_h) & \text{if } \theta_h \geq x^{-\frac{\alpha}{2}}, \\ f_1(x^{-\frac{\alpha}{2}}) & \text{otherwise} \end{cases}, \quad (49)$$

and

$$C_2 = \begin{cases} f_2(\theta_l) & \text{if } \theta_l \leq x^{-\frac{\alpha}{2}}, \\ f_2(x^{-\frac{\alpha}{2}}) & \text{otherwise,} \end{cases} \quad (50)$$

respectively, where θ_h and θ_l are determined by (31) and (32), respectively. It is clear that $f_1(\theta_h) \geq f_1(x^{-\frac{\alpha}{2}})$ when $\theta_h \geq x^{-\frac{\alpha}{2}}$, and $f_2(\theta_l) \geq f_2(x^{-\frac{\alpha}{2}})$ when $\theta_l \leq x^{-\frac{\alpha}{2}}$. According to (31) and (32), if $\theta_h = x^{-\frac{\alpha}{2}}$, then we have $\theta_l = \theta_h$. In this case, we have

$$f_1(\theta_h) = f_1(x^{-\frac{\alpha}{2}}) = f_2(\theta_l) = f_2(x^{-\frac{\alpha}{2}}). \quad (51)$$

In the following, we show that if $\theta_h > x^{-\frac{\alpha}{2}}$, then we have $\theta_l > x^{-\frac{\alpha}{2}}$ by contradiction. In particular, if $\theta_l \leq x^{-\frac{\alpha}{2}}$, then

$$\begin{aligned} \frac{\theta_l}{1 + \theta_l} &= \ln(1 + \theta_l) \cdot \left(\frac{2}{\alpha} \cdot x \cdot \theta_l^{\frac{2}{\alpha}} + R^\alpha \sigma^2 \theta_l \right) \\ &\leq \ln(1 + \theta_l) \cdot \left(\frac{2}{\alpha} + R^\alpha \sigma^2 \theta_l \right), \end{aligned} \quad (52)$$

which indicates that $\theta_l \geq \theta_h$. With $\theta_h > x^{-\frac{\alpha}{2}}$, we have $\theta_l > x^{-\frac{\alpha}{2}}$, which results in contradiction. Similarly, it can be shown by contradiction that if $\theta_h < x^{-\frac{\alpha}{2}}$, then we have $\theta_l < x^{-\frac{\alpha}{2}}$.

Therefore, when $\theta_h \geq x^{-\frac{\alpha}{2}}$, we have

$$C_1 = f_1(\theta_h) \geq f_1(x^{-\frac{\alpha}{2}}) = f_2(x^{-\frac{\alpha}{2}}) = C_2. \quad (53)$$

We then have $C_{\text{max}} = f_1(\theta_h)$. Similarly, when $\theta_h < x^{-\frac{\alpha}{2}}$, we have $C_{\text{max}} = f_2(\theta_l)$. By $\theta_h = x^{-\frac{\alpha}{2}}$, we have

$$\sigma^2 = \left(\frac{1}{(1 + x^{\frac{\alpha}{2}}) \ln(1 + x^{-\frac{\alpha}{2}})} - \frac{2}{\alpha} \right) R^{-\alpha} x^{\frac{\alpha}{2}}. \quad (54)$$

Therefore, $\theta_h \geq x^{-\frac{\alpha}{2}}$ is equivalent to

$$\sigma^2 \leq \left(\frac{1}{(1 + x^{\frac{\alpha}{2}}) \ln(1 + x^{-\frac{\alpha}{2}})} - \frac{2}{\alpha} \right) R^{-\alpha} x^{\frac{\alpha}{2}}, \quad (55)$$

which is shown in (28).